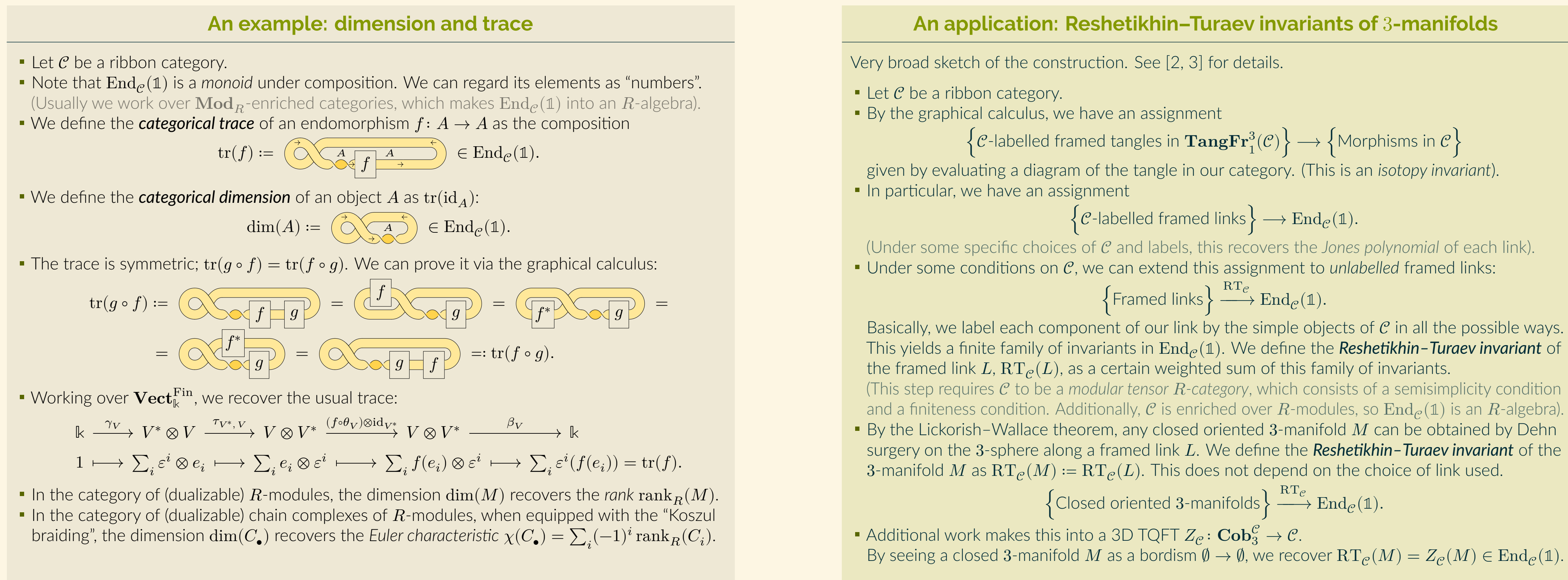


Santiago Pareja Pérez



- Similar diagrams, but we color the regions to denote 0-cells (objects).
- Consider the bicategory **Cat** of (small) categories, functors, and natural transformations.

This diagram depicts one half of the definition of adjointness of functors $\mathcal{C} \overset{L}{\underset{R}{\rightleftarrows}} \mathcal{D}$.

- $\mathcal{C}$ and $\mathcal{D}$ are categories (objects).
- $L: \mathcal{C} \rightarrow \mathcal{D}$ and $R: \mathcal{D} \rightarrow \mathcal{C}$ are functors (1-morphisms).
- The unit $\eta: \text{id}_{\mathcal{C}} \Rightarrow R \circ L$ and counit $\nu: L \circ R \Rightarrow \text{id}_{\mathcal{D}}$ are natural transformations (2-morphisms).

